Summer Ann

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Education

University of Chicago Sep 2024 – Aug 2027
Doctor of Philosophy – PhD, Computer Science *GPA: 3.98, Full Ride Scholarship*

- Focus Areas: Machine Learning, Large Language Models, Computational Medicine, Reinforcement Learning
- Developed pipelines for immune receptor data and processed large-scale datasets for AI-ready analysis
- Relevant Coursework: Advanced Machine Learning, Statistical Learning Theory, Quantum Computing Systems, Optimization, Transformers, Computational Biology

University of Michigan Aug 2021 – May 2024
Bachelor of Science – BS, Biology, Computer Science, University Honors

- Activities: NSA, Michigan Health Aid, Detroit Partnership Planning Team, UMARV
- TA for EECS 548/SI 642, guided graduate students in Python-based ML data pipelines
- Relevant Coursework: Programming, Mobile App Development, NLP, Data Management

Publications and Accepted Papers

The Interaction Tax: Communication Erases Diversity in Multi-Agent Teams Accepted, 2026
ICML 2026 Workshop AI4Research

- Accepted workshop paper studying how communication can erase diversity in multi-agent teams and affect collaborative AI-for-research systems.
- Analyzes the “interaction tax” in agent coordination, with implications for scientific reasoning, multi-agent evaluation, and robust agent-team design.

Professional Experience

Co-Founder Jan 2025 – Present
DevSecCode, Inc. *Chicago, IL, Remote*

- Architected Deva, an AI-native secure coding and compliance platform that analyzes code during development, detects insecure patterns, explains risk, and generates context-aware remediations before code reaches review or audit.
- Grew Deva to 20,000+ users within three weeks of public launch across the Deva IDE, VS Code Marketplace, Open VSX Registry, and npm distribution, after earlier traction of 9,500+ npm downloads and 1,900+ extension downloads at launch.
- Designed the core security engine across layered rule-based scanning, AST-aware code analysis, local reachability analysis, dependency and dataflow enrichment, calibrated confidence metadata, and advisory-aware and suppression-aware filters.
- Built compliance-aware workflows mapping findings to HIPAA, SOC 2, PCI DSS, NIST 800-53, CWE/CVE classes, SARIF, and OSCAL outputs, with Deva Hospital, Deva Gov, and Deva Finance model tracks.
- Secured validation through AWS Activate, NVIDIA Inception, MACH37, Blu Ventures Cyber Venture Forum 2026, Techstars pitch series, and design partner conversations across healthcare, government cyber/intelligence, finance, and defense technology. Skills: Application Security, Secure Code Analysis, LLM Systems, Compliance Automation, Developer Tooling

Co-CTO Jun 2025 – Present
Concord Systems Corp. *New York, NY, Remote*

- Spearheaded integration of Fireblocks’ MPC approval flow, reducing end-to-end approval latency by ~40% and cutting failed-signature incidents by 25%.
- Designed trade execution types and predictive models that reduced simulated frontrunning and adverse selection by 18% on historical order book replays.
- Developed scripts for real-time MPC transaction coordination, increasing system throughput from ~50 to 200+ signed transactions per second in internal benchmarks.
- Skills: Trading Systems, Quantitative Research, Web3, Machine Learning

LLM/Data Engineer May 2024 – Present
Neuromuse.ai *Ann Arbor, MI, Remote*

- Led integration of instruction-tuned LLMs into a natural language-to-music generation stack, designing a modular prompting and postprocessing layer for style, tempo, and instrumentation control.
- Built evaluation pipelines combining automatic metrics such as semantic similarity and lexical overlap with human preference studies, increasing “on-brief” generations from 72% to 88%.
- Implemented dataset curation and deduplication for text-audio pairs, reducing training set noise by ~30% and improving downstream alignment scores by 5–7 points.
- Prototyped retrieval-augmented workflows that condition music prompts on user libraries, enabling creator-specific style transfer without retraining the base LLM.
- Skills: Large Language Models, Artificial Intelligence, Product Management

Internships

AI/ML Solutions Architect Intern May 2024 – Sep 2024
Amazon Web Services (AWS) *Seoul, South Korea*

- Designed and implemented “ZON,” an LLM-powered scheduling assistant on AWS Bedrock that performs multi-step calendar reasoning and constraint satisfaction across timezones.
- Compared prompt engineering vs. toolformer-style function calling for calendar operations, reducing failure cases such as invalid timeslots and missing constraints by 30%.
- Built observability and feedback loops with structured logs, telemetry, and user feedback signals to iteratively refine prompts and guardrails, cutting hallucinated events by ~40%.
- Deployed the system on a scalable EC2-based architecture with OAuth 2.0, supporting 10k+ concurrent sessions in load tests with <200ms median LLM orchestration latency.
- Skills: Machine Learning, LangChain, Cloud Computing, LLM

Software Engineering Fellow Jun 2023 – Aug 2023
Backyard Brains *Belgrade, Serbia*

- Performed Pandas-based EEG data analysis, decreasing data cleaning and feature extraction time by ~50% for neuroscience experiments.
- Developed eyetracker computer vision-based games that doubled experimental task engagement compared to a non-interactive baseline.
- Delivered prototypes later showcased on TED Talks, helping the team reach thousands of students and educators.
- Skills: Computer Vision, Flask, Python, Node.js, Flutter

Research Experience

PhD Researcher Sep 2024 – Present
University of Chicago, Data Commons / Grossman Lab *Chicago, IL*
- Conduct research on AI-driven immune receptor modeling and biomedical machine learning pipelines for large-scale immune receptor datasets.
- Develop data processing, quality control, and analysis workflows that support AI-ready biomedical datasets and downstream computational medicine research.
- Contribute to reproducible research infrastructure for large biomedical datasets, with emphasis on scalable analysis, data quality, and translational impact.
- Skills: Computational Medicine, Biomedical ML, Data Pipelines, Immune Receptor Modeling

Researcher Sep 2024 – Present
University of Chicago, Chenhao Tan Group *Chicago, IL*
- Conduct research on multi-agent systems for scientific reasoning, with emphasis on agent coordination, long-horizon execution, and evaluation under realistic compute and validity constraints.
- Built infrastructure for execution-grounded benchmarking, automated validity checks, and large-scale auditing of LLM-generated scientific workflows and experiment runs.
- Designed agent architectures and persistent runtime systems for long-running scientific workflows, including structured communication, memory, provenance, and reproducibility mechanisms.
- Skills: Multi-Agent Systems, Scientific AI, LLM Evaluation, Systems for ML

Researcher Jun 2024 – Present
University of Chicago, Beaulieu-Jones Lab *Chicago, IL*
- Conduct research on interpretable deep learning for ECG classification, with emphasis on prototype-based architectures, reproducibility, and clinically meaningful model interpretation.
- Re-engineered training and evaluation pipelines for PTB-XL and EchoNext, enabling reproducible experiments with versioned configurations, seeded runs, and provenance tracking.
- Built analysis tooling for prototype activation maps, lead-wise relevance, and nearest-neighbor retrieval to support expert inspection of learned representations and diagnostic behavior.
- Skills: Medical AI, Explainable AI, Deep Learning, PyTorch

Undergraduate Researcher Sep 2022 – Sep 2024
Michigan Medicine, Boyle Lab *Ann Arbor, MI*
- Created ML pipelines for genomic STR analysis, improving classification performance by up to 9% AUROC over baseline feature-engineered models.
- Explored STR subclasses for disease analysis using clustering and dimensionality reduction, revealing candidate subgroups in multiple disease cohorts used to refine follow-up experiments.
- Automated data preprocessing and quality control, reducing manual curation time per dataset from hours to under 10 minutes.
- Skills: Bioinformatics, Deep Learning, Computational Biology

Selected Research Projects

Agent4Science: Multi-Agent Platform for Scientific Collaboration Sep 2024 – Present
University of Chicago
- Built a multi-agent platform for scientific collaboration in which specialized agents for literature retrieval, hypothesis generation, experiment design, and analysis coordinate on end-to-end research workflows.
- Designed role-specific agent architectures and structured communication protocols enabling compositional task decomposition, delegated execution, and persistent multi-step research sessions.
- Integrated the platform with persistent runtime and evaluation infrastructure to support long-horizon workflows, execution-grounded validation, and reproducible benchmarking.
- Skills: Multi-Agent Systems, LLM Orchestration, Scientific AI, Agent Infrastructure

Marginal Epistemic Gain (MEG): Compute-Matched Evaluation of Multi-Agent Research Mar 2026 – Present
University of Chicago, NeurIPS 2026 in preparation
- Proposing MEG as a principled metric for evaluating multi-agent scientific systems under fixed compute budgets, measuring the incremental knowledge contributed per unit of inference cost.
- Developing experimental protocols to compare multi-agent collaboration strategies such as parallel search, sequential refinement, and debate against single-agent baselines at matched compute.
- Skills: LLM Evaluation, Multi-Agent Systems, Experimental Design, Scientific AI

HypoGen / HypoBench FailureLab: Execution-Grounded AI Research Validation Jan 2026 – Present
University of Chicago
- Built automated validity checks and indexing workflows to evaluate large batches of LLM-generated research runs in an execution-grounded environment.
- Implemented run aggregation and auditing pipelines, generating a master index over 1,728 experiment runs and producing dataset-level summary statistics for completeness and failure modes.
- Analyzed failure patterns such as diversity collapse behaviors to inform more robust search and training strategies.
- Skills: LLM Evaluation, Experimentation Systems, Reproducibility, Python

Flamebird: Persistent Agent Runtime Jan 2025 – Present
University of Chicago
- Building runtime infrastructure for long-lived, stateful AI agents with persistent memory, durable task queues, and fault-tolerant execution across sessions.
- Implementing agent lifecycle management including spawn, checkpoint, resume, and retire with pluggable backends for state persistence and inter-agent messaging.
- Designing the system to serve as the execution backbone for Agent4Science, supporting multi-day research workflows with provenance tracking and reproducibility guarantees.

- Skills: Systems Engineering, Agent Infrastructure, Distributed Systems, Python

CLMD: Compressive Long-Term Memory for Dialogue Agents Sep 2024 – Present
University of Chicago
- Designed a dual-memory architecture that combines a short-term scratchpad with compressed "memory cards," reducing average context length by up to 45% at fixed task accuracy.
- Implemented FAISS- and SQLite-backed memory stores with pluggable retrieval policies including BM25, dense embeddings, and hybrid search, sustaining sub-50ms retrieval on 100k+ memories.
- Conducted ablations over write policies, showing ~20% reduction in explicit self-contradictions relative to a context-only baseline.
- Skills: Large Language Models, Retrieval-Augmented Generation, Systems for ML, Python

Projects

Deva: AI-Native Secure Coding and Compliance Platform Jan 2025 – Present
DevSecCode, Inc.
- Built an editor-centered platform that scans code as it is written, surfaces security and compliance issues early, and generates developer-facing explanations and remediations before downstream review.
- Developed a layered architecture spanning static analysis, policy and compliance mapping, vulnerability triage, and LLM-assisted fix generation for engineering teams building in regulated environments.
- Designed product workflows for secure code generation, code review assistance, and requirement-aware remediation across application security and compliance use cases.
- Skills: AppSec, Static Analysis, LLM Product Engineering, Compliance Systems

Product and Application Portfolio Jan 2024 – Present
Summit Wanderlust, LLC and Open Source
- Yammoing: iOS platform for scanning and analyzing food, water, cosmetics, and home goods with structured parsing, category-specific safety insights, and local-first data management.
- BreatheMindful: SwiftUI breathing, focus, and sleep companion with Apple HealthKit integration; reached 150+ beta users in the first month with a 32% increase in weekly session completion.
- Lovocado (CouplesConnect): Private, local-only couples app with daily check-ins, care task tracking, shared notes, trip planning, and HealthKit-synced fitness challenges, all stored on-device with no servers or accounts.
- My Little Chef: Premium cooking experience treating recipes as cinematic, collectible media with voice-led cook mode, smart timers, mood-based discovery, and a remix/forking culture.
- MyPlanner: Cross-platform planner suite for iOS and macOS with Notion-inspired UX, NLP-based scheduling, and AI-assisted task organization.
- CreatorFlow AI: Dockerized local LLM stack for content creators with RAG pipelines, A/B prompt testing across platforms, and per-request telemetry; boosted estimated hook CTR by 18%.
- With You: Full-stack university community platform built with Next.js 14, Prisma, and NextAuth, featuring timetable scheduling, course reviews, forums, marketplace, and university-email-gated membership.
- CashCart: Grocery price comparison app shipped to the App Store with 500+ downloads; Flask backend with <150ms median API response time.
- Skills: SwiftUI, Next.js, HealthKit, Docker, RAG, Full-Stack Development, Product Design

Transformer-Based Financial Text Analysis Feb 2025 – Feb 2025
University of Chicago
- Implemented a GPT-2-like transformer with einops for financial text analysis, outperforming a strong logistic regression baseline by 14% F1 on earnings sentiment.
- Analyzed earnings reports for sentiment prediction using PyTorch, improving drawdown-aware trading PnL by ~8% in backtests.
- Skills: Large Language Models, PyTorch, Quantitative Finance

Domain Adaptation of LLMs for Cancer Classification Feb 2025 – Feb 2025
University of Chicago
- Compared general-purpose BERT vs. PubMedBERT embeddings on TCGA-derived text features, with domain-specific pretraining improving macro-F1 by 11% on multi-class cancer prediction.
- Built a clean evaluation pipeline with stratified splits, nested cross-validation, and calibration curves to quantify the effect of domain adaptation on both accuracy and confidence calibration.
- Skills: Large Language Models, Hugging Face, Bioinformatics

Multithreaded Stock Trading System Jan 2025 – Mar 2025

- Developed a high-performance C++20 backend simulating a stock trading environment capable of processing 100k+ orders per second on a single commodity machine.
- Modeled traders, order books, and matching engines with multithreading, demonstrating near-linear scaling up to 8 threads in synthetic benchmarks.
- Skills: C++20, Multithreading, Financial Modeling, Quantitative Finance

TCR-GraphDiff for Immune Receptor Analysis Jan 2025 – Mar 2025
University of Chicago
- Developed a graph neural network for TCR sequencing analysis, improving predictive performance on held-out cohorts by ~7% AUROC vs. non-graph baselines.
- Curated datasets for Immune Receptor Data Commons, standardizing metadata and improving downstream pipeline reliability.
- Skills: Graph Neural Networks, Deep Learning, Bioinformatics

LLM Probing for Code Semantics vs. Syntax Jan 2025 – Mar 2025
University of Chicago
- Constructed probing datasets from Leetcode-Rosetta and AST-preserving vs. semantics-altering transformations to disentangle syntactic and semantic information in LLM code representations.
- Designed a hybrid subspace-probing framework that outperformed naive linear probes by 9% accuracy on semantic discrimination tasks while controlling for superficial syntax cues.
- Skills: Large Language Models, Hugging Face, Mechanistic Interpretability

RL-Based Trading Strategy Optimization Jan 2025 – Mar 2025
University of Chicago
- Designed RL models using OpenAI Gym for trading strategies, with Q-learning and DQN policies achieving 10% improved portfolio returns

in backtests over a buy-and-hold baseline.
- Conducted ablation studies on state representations and reward shaping, reducing maximum drawdown by ~12% versus naive RL configurations.
- Skills: Reinforcement Learning, Quantitative Finance, Python

Stock Price Movement Prediction using SVM, KRR Oct 2024 – Dec 2024
University of Chicago
- Developed SVM and KRR models achieving F1-score of 0.80, improving directional hit-rate by 13% over a logistic regression baseline.
- Applied K-Means to identify 31 stock groupings, improving risk-adjusted returns in a sector-tilted portfolio simulation by 7%.
- Skills: SVM, Cluster Analysis, Quantitative Analysis

Explainable AI Tools for Education Jan 2023 – Present
University of Michigan
- Co-authored a study on explainable AI for education, demonstrating 20% higher concept retention when students interacted with model explanations.
- Developed frameworks for transparent ML models used in classroom pilots with 100+ students.
- Skills: Machine Learning, Explainable AI, Python

Hummingbot (High Frequency Trading) Dec 2023 – Dec 2023
University of Michigan
- Contributed to Hummingbot exchange connectors, reducing order submission latency by ~25% on testnet environments.
- Developed V2 strategies for multi-venue trading that improved simulated market-making PnL by 6% over existing templates.
- Skills: Quantitative Analysis, Java, Python, Algorithmic Trading

Skills

Languages – Proficient (5+ Years): Python, C++, SQL, R, MATLAB

Languages – Intermediate (2+ Years): Java, JavaScript, Swift

Primary ML Frameworks and Libraries: PyTorch, PyTorch Geometric, PyTorch Lightning, Optuna, NetworkX, NumPy, Scikit-learn, TensorFlow, LangChain, Hugging Face, OpenAI Gym, Qiskit, einops

AWS Expertise: SageMaker, Lambda, Bedrock

Languages

Korean (Native), English (Native), German (Professional/C1)